Class 15: Encoding and Embedding Practice

Represent the categories, inspect the result, and state what the evidence supports

Data-driven Systems Engineering

23 October

Attributes in the prepared Titanic dataset

Attribute	Description	Role in this activity
sex	Passenger sex recorded in the dataset	Nominal feature
pclass	Ticket class: 1st, 2nd, or 3rd	Ordered feature
embarked	Port of embarkation: C = Cherbourg, Q = Queenstown, S = Southampton	Nominal feature
age	Passenger age in years; missing values already imputed	Numeric feature
fare	Passenger fare	Numeric feature
sibsp	Number of siblings and spouses aboard	Numeric feature
parch	Number of parents and children aboard	Numeric feature
survived	Outcome: 0 = did not survive; 1 = survived	Target

Scope of the supplied CSV

This prepared file contains eight passenger attributes for all 891 passengers. It is a selected, imputed subset for this exploratory activity; other fields from the full Titanic dataset are not included.

Today's question: is this a useful representation?

1. Build encodings for Titanic categories and inspect the resulting feature matrix.
2. Explore how the categories relate to survival, without fitting a predictive model.
3. Construct a PCA projection of passenger profiles grouped by embarkation port.
4. Separate evidence about representation quality from evidence about model contribution.

Prepared data

Use all 891 rows in the prepared Titanic dataset for today's encoding and association explorations. No predictive model is evaluated here, so a train/test split is unnecessary. Imputation choices are the focus of Class 16. In Class 18, create the split before fitting any preprocessing for model evaluation.

Download the prepared Titanic dataset (CSV)

Pick a representation that matches the category

Feature	Structure	Representation to inspect
sex	Nominal, two categories	One-hot; no natural order
embarked	Nominal, three ports	One-hot; check unseen-category behavior
pclass	Ordered categories	Compare one-hot with ordinal codes $1, 2, 3$

- Ordinal codes preserve order but add a distance assumption: the step from 1 to 2 equals the step from 2 to 3.
- One-hot encoding preserves category identity without adding an order.
- A learned or PCA-based embedding assigns coordinates; what those coordinates mean depends on how they were learned.

One-hot turns categories into explicit columns

Passenger	Input categories `sex`	`port`	Output: sex female	male	Output: embarked C	Q	S
A	female	C	**1**	0	**1**	0	0
B	male	S	0	**1**	0	0	**1**
C	female	Q	**1**	0	0	**1**	0

What changed?

One category becomes one column. The active column records the observed category; no artificial order is introduced.

What to check?

Fit columns on the activity dataset. Confirm expected levels and document how an unseen value is handled at transform time.

First evaluate representation quality

Validity — Are the levels preserved?

How `enc.get_feature_names_out()`

Evidence Learned categories and output column names

Says Expected levels are preserved

+ Good `sex_female, sex_male, embarked_C/Q/S`

- Bad A missing level, or a stray `embarked_nan` column

Robustness — Does an unseen value break it?

How `OneHotEncoder(handle_unknown="ignore")`

Evidence Transform a row with an unseen value

Says Behavior is documented

+ Good Unseen port `"X"` becomes an all-zero block, no error

- Bad Crash at transform time, or the behavior is never checked

Complexity — Is the size manageable?

How `Z.shape; Z.nnz / (n * p)`

Evidence Output columns and matrix density

Says Size and sparsity are manageable

+ Good 8 columns for 3 features, mostly zeros

- Bad An ID-like column with 891 levels gives 891 columns

Exploratory signal — Do groups differ?

How `df.groupby("embarked").agg(...)`

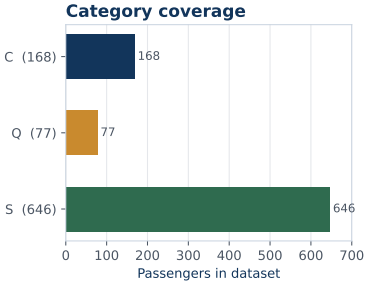
Evidence Category counts and survival rates per group

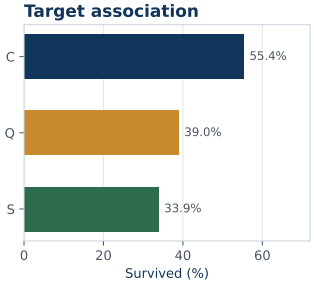
Says Group differences suggest a hypothesis

+ Good C: 168 passengers, 55% survived; S: 646, 34%. Report counts with rates

- Bad Claiming Q improves a model from a 77-passenger rate

Check category coverage and compare group rates





All 891 passengers · descriptive rates are not model-performance evidence

Read the chart

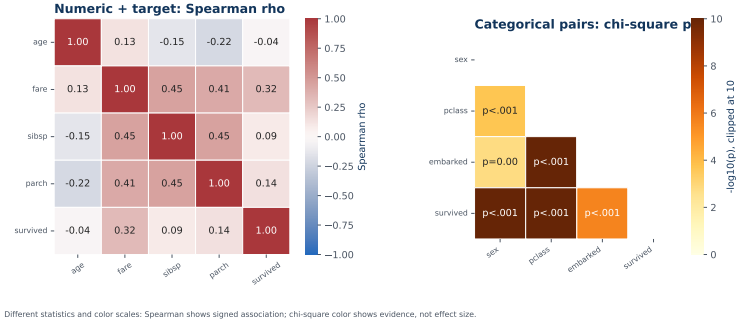
All ports appear in the dataset, but their counts differ. Compare the survival rates while keeping those sample sizes in view.

Quality check Exploratory signal ▪ Validity

+ **Good** All three ports present; counts shown beside rates (C 168, Q 77, S 646). Differences are read as a hypothesis.

- **Bad** Comparing rates without sample sizes, or claiming the encoding improves held-out predictions.

Association heatmaps: read each statistic on its own scale



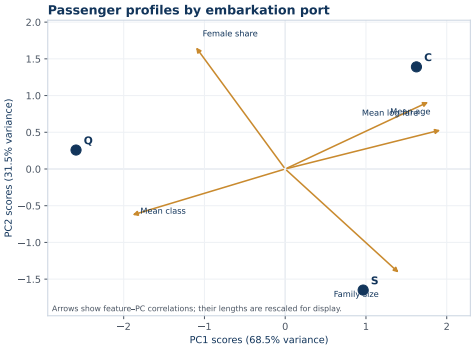
Different statistics and color scales: Spearman shows signed association; chi-square color shows evidence, not effect size.

Quality check Exploratory signal

+ Good Spearman ρ read as signed rank association; χ^2 colors read as evidence against independence (p-values), each on its own scale.

- Bad Comparing colors across panels, or treating a small p-value as effect size.

PCA shows how original features align with the profile axes



- Points: three port profiles, built without `survived`.
- Arrows: correlations of original features with PC1 and PC2.
- Similar directions suggest features that vary together in these profiles.

Quality check Complexity ▪ Exploratory signal

+ **Good** Profiles built without `survived`; arrows read as how features align with PC1 and PC2.

- **Bad** Reading "two PCs explain 100%" as predictive value: with three centered profiles it is guaranteed.

One-hot PCA is not a useful port embedding here

- Three one-hot port columns have at most two independent directions after centering.
- PCA to two components therefore keeps all centered variation; it mostly rotates the same information.
- The resulting coordinates do not learn that two ports are behaviorally or predictively similar.
- Use the profile PCA as a small visualization exercise; keep one-hot as the clear baseline representation.

Representation quality today

Check that categories are preserved, the matrix is manageable, unknown values are handled, and any visual or target association is interpreted within its limits.

Can we say encoding contributed to modeling? Not yet.

We can assess today

- Correctness and category coverage
- Unknown-category handling
- Number and density of features
- Descriptive association with the target
- Structure of the PCA profile plot

We cannot claim today

- Better generalization
- Higher predictive performance
- That an embedding beats one-hot
- That target association adds value alongside other features

Conclusion

Encoding changes how a model receives information; it does not create new information by itself. Measuring predictive contribution requires a downstream model.

How Class 18 will test contribution fairly

1. Create one stratified train/test split and keep it fixed across the comparisons.
2. Compare a baseline without the feature, one-hot `embarked`, and any candidate embedding.
3. Fit every learned transformation on training data only; evaluate once on the held-out test data. Start from raw data, not this all-rows imputed activity fixture.
4. Choose a suitable metric in advance (e.g., ROC AUC or balanced accuracy) and compare the change, not just one score.
5. Repeat across folds or seeds if time permits; report uncertainty and added complexity.

Today's deliverable

A representation table, a short association interpretation, the PCA profile plot, and one explicit claim the evidence does *not* support yet.

Debrief and transition

- Which encoding assumptions were reasonable for `sex`, `embarked`, and `pclass`?
- Did the PCA plot describe category profiles, or did it prove predictive usefulness?
- Which representation would you carry forward as a baseline, and why?

Next: Class 16

Feature engineering returns to the raw data and examines cleaning, curation, missing values, construction, and selection. The prepared table today keeps that imputation discussion in its proper session.